Computer Architecture

Discussion Session V

Prof. Onur Mutlu
ETH Zürich

Fall 2025

14 December 2025

Agenda for the Discussion Session (1/3)

- Preparation of Final Exam – Fall 2024
  - HW1 Q4.2 from Fall 2022 - In-DRAM Bitmap Indices
    - Link to Question PDF: https://safari.ethz.ch/architecture/fall2022/lib/exe/fetch.php?media=hw1.pdf
    - Link to YouTube Solution: https://www.youtube.com/watch?v=aFeJv9tQ3-4&t=0s
  - HW3 Q5 from Fall 2022 – BossMem
    - Link to Question PDF: https://safari.ethz.ch/architecture/fall2022/lib/exe/fetch.php?media=hw3.pdf
    - Link to YouTube Solution: https://www.youtube.com/watch?v=aFeJv9tQ3-4&t=1415s
  - HW4 Q6 from Fall 2021 - Cache Coherence
    - Link to Question PDF: https://safari.ethz.ch/architecture/fall2021/lib/exe/fetch.php?media=hw4.pdf
    - Link to YouTube Solution: https://www.youtube.com/watch?v=aFeJv9tQ3-4&t=2512s

Agenda for the Discussion Session (2/3)

- Preparation of Final Exam – Fall 2024
  - HW4 Q4 from Fall 2022 - Runahead Execution
    - Link to Question PDF: https://safari.ethz.ch/architecture/fall2022/lib/exe/fetch.php?media=hw4.pdf
    - Link to YouTube Solution: https://www.youtube.com/watch?v=aFeJv9tQ3-4&t=3697s
  - HW5 Q8 from Fall 2020 - Building Multicore Processors
    - Link to Question PDF: https://safari.ethz.ch/architecture/fall2020/lib/exe/fetch.php?media=hw_5-v2.pdf
    - Link to YouTube Solution: https://www.youtube.com/watch?v=aFeJv9tQ3-4&t=4654s
  - HW5 Q5 from Fall 2020 – Prefetching
    - Link to Question PDF: https://safari.ethz.ch/architecture/fall2020/lib/exe/fetch.php?media=hw_5-v2.pdf
    - Link to YouTube Solution: https://www.youtube.com/watch?v=aFeJv9tQ3-4&t=5392s

Agenda for the Discussion Session (3/3)

- Preparation of Final Exam – Fall 2024
  - HW3 Q3 from Fall 2020 - Genome Analysis
    - Link to Question PDF: https://safari.ethz.ch/architecture/fall2020/lib/exe/fetch.php?media=hw_3.pdf
    - Link to YouTube Solution: https://www.youtube.com/watch?v=aFeJv9tQ3-4&t=5782s
  - HW4 Q6 from Fall 2022 - Memory Consistency
    - Link to Question PDF: https://safari.ethz.ch/architecture/fall2022/lib/exe/fetch.php?media=hw4.pdf
    - Link to YouTube Solution: https://www.youtube.com/watch?v=aFeJv9tQ3-4&t=7930s

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